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Worksheet-9

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Aim of the Session:

The aim of this practical session is to implement database triggers in PostgreSQL to automatically calculate values and enforce constraints during data insertion operations.

Objective of the Session:

The objectives of this practical session are:

- To understand the concept of triggers in DBMS.
- To automate actions using triggers during INSERT operations.
- To enforce constraints and maintain data integrity.
- To perform automatic calculations using triggers.
- To reduce manual intervention in database operations.

Practical / Experiment Steps:

Step 1: Creating a table for demonstration.

- Implementing a table to store employee salary details.

Step 2: Creating a Trigger Function.

- Creating a function to calculate bonus automatically.

Step 3: Creating a Trigger.

- Creating a trigger that executes before inserting data.

Step 4: Inserting Data.

- Inserting records to check automatic trigger execution.

Step 5: Verifying Results.

- Displaying table data to confirm trigger functionality.

Procedure of the Practical:

1. Start the system and log in to the computer.
2. Open the PostgreSQL client tool (psql / pgAdmin).
3. Create Tables.

```
CREATE TABLE EmployeeSalary (
   EmpID SERIAL PRIMARY KEY,
   EmpName VARCHAR(50),
   Salary DECIMAL(10,2),
   Bonus DECIMAL(10,2)
);
```

4. Create Trigger Function

```
CREATE OR REPLACE FUNCTION calculate_bonus()
RETURNS TRIGGER AS $$
BEGIN
   NEW.Bonus := NEW.Salary * 0.10;
   RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

5. Create Trigger

```
CREATE TRIGGER before_insert_bonus
BEFORE INSERT ON EmployeeSalary
FOR EACH ROW
EXECUTE FUNCTION calculate_bonus();
```

6. Insert Data

```
INSERT INTO EmployeeSalary (EmpName, Salary)
VALUES ('Amit', 50000),
('Riya', 60000);
```

7. Display Data

```
SELECT * FROM EmployeeSalary;
```

Data Output Messages Notifications

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	empid [PK] integer	empname character varying (50)	salary numeric (10,2)	bonus numeric (10,2)
1	1	Amit	50000.00	5000.00
2	2	Riya	60000.00	6000.00

8. Verify the output after execution.
9. Note down the results obtained.
10. Save the work and take screenshots for record.

I/O Analysis:

Input Provided:

- Table structure for employee salary
- Trigger function definition
- Trigger creation query
- Insert statements

Output Generated:

- Automatic calculation of bonus (10% of salary)
- Inserted records with computed bonus values
- Maintained data consistency

Learning Outcomes:

- Understanding the concept of database triggers
- Ability to automate database operations
- Enforcing constraints using triggers
- Reducing manual effort through automation
- Practical knowledge of PostgreSQL trigger implementation